

Solution Requirements

Date	30 September 2026
Team ID	SWTID-2026-8026
Project Name	Import Data Using Transform Maps (Spreadsheet)
Team Size	3
Team Leader	Santhakumari V
Team Members	Nalayiram Chinnaveni A, Seema M
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users log in to the ServiceNow instance with their own username and password (or company SSO). Only logged-in users can open the import features.	High
2	Authorization levels	Admin: can upload files, create data sources and transform maps, and run imports. Import user: can upload files and run existing maps. Business user: can only view the imported records in the target table.	High
3	External interfaces	Reads Excel (.xlsx) and CSV files uploaded from the user's computer. Uses the ServiceNow Import Set and Transform Map tables. Uses the email server to send the completion notification.	Medium

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
4	Transactions processing	Rows are first loaded into the import set (staging) table. The transform map then inserts or updates records in the target table. A coalesce field avoids duplicates, and failed rows are skipped and logged without stopping the whole import.	High
5	Reporting	Import summary showing the number of inserted, updated, ignored and error rows. Import log / transform history. Imported records can be viewed in a list and exported.	Medium
6	Business rules, etc.	Only .xlsx and .csv files are accepted. Required columns must be present. Every transform map must have a coalesce field. Field scripts convert dates and choice values before saving.	High
7	Compliance to laws or regulations	Data is protected with the platform's access controls (ACLs). Personal data in the spreadsheets is visible only to permitted roles and handled as per the organisation's data privacy policy (e.g., GDPR).	Low
8	Other	The same transform map can be reused for spreadsheets with the same format. An email is sent when an import is completed.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	A file of around 5,000 rows is loaded and transformed in a few minutes. Bigger files are run as a background job so the instance does not slow down.	Up to 5,000 rows in less than 5 minutes
2	Scalability	Larger files are loaded in batches, and new transform maps can be added for other tables later.	Supports files up to 20,000 rows per import

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
3	Security & Data Privacy	Uploaded data is protected by role-based access and ACLs, and all traffic goes over HTTPS.	Only users with import roles can upload or run imports
4	Reliability & Availability	One bad row must not stop the whole import, and an interrupted import can be run again safely.	99% uptime per month; no duplicate records
5	Usability & Accessibility	The steps (upload, map, transform) are simple and error messages are clear, so an admin can finish an import without special training.	First import completed without any help
6	Other	Maintainability: transform maps are reusable and field names are kept clear.	Same map works for new files of the same format